

DOCKING PIPELINE USER GUIDE

OVERVIEW

This pipeline performs automated ligand docking using AutoDock Vina.

It:

- downloads or uses a receptor
- prepares ligands
- performs docking
- outputs ranked binding energies
- optionally prepares structures for visualization

SETUP

Activate environment:

```
conda activate docking_env
```

Navigate to pipeline directory:

```
cd ~/docking
```

INPUT FILES

Place ligand files in:

ligands/

Supported formats:

.sdf, .mol, .mol2

CONFIGURATION (docking_config.yml)

Receptor:

pdb_id: 3EST

receptor: "" (leave blank to download)

Docking box (REQUIRED):

center_x, center_y, center_z (from PDB)

size_x, size_y, size_z (Å)

RUNNING

python3 run_pipeline.py docking_config.yml

OUTPUT

runs/docking_run_TIMESTAMP/

results/

binding_scores.csv

top_ligands.txt

INTERPRETATION

< 0 favorable

~ -4 weak

~ -7 moderate

< -10 strong

> 0 failed

VISUALIZATION (optional)

chimerax @/full/path/to/view_top_ligands.cxc

(only negative-energy ligands included)

NOTES

- Use heavy atom coordinates
- Each run is reproducible